

Gyaneshwar Agrahari

Ph.D. Candidate in Mathematics

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Research Interests

Structural and Extremal Graph Theory, Math Education, Graph-based Machine Learning, Matroid Theory, Geospatial Analysis

Education

Ph.D. in Mathematics , Louisiana State University	2022 – Present
Advisor: Zhiyu Wang	GPA-4.0
Expected Graduation: May 2027	
<i>Thesis: Counting Subgraphs in Planar Graphs with Higher Connectivity</i>	
B.S. in Mathematics, B.S. in Physics , Youngstown State University	2018 – 2022
Graduated with Summa Cum Laude, Honors	GPA-3.9

Publications & Preprints

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1. G. Agrahari and Z. Wang, *On the number of $K_{1,t}$ and $K_{2,t}$ in 4 or 5-connected planar triangulations*, In preparation.
 2. G. Agrahari, C. Bibby, S. Boros, H. Garcia, F. Heidercheit, and Z. Wang *On some structural properties of graphs with non-negative resistance curvature*, In preparation.
 3. G. Agrahari, K. Bist, M. Pandey, J. Kapita, Z. James, J. Knox, S. Ramirez, S. Heymsfield, P. Wolenski, and N. Drenska, *Predicting Biometrics Using Supervised and Semi-Supervised Algorithms on Anthropometric Measurements*, *Frontiers in Bioinformatics*, vol. 6, 2026, doi:10.3389/fbinf.2026.17225
 4. G. Agrahari, X. Liu, Z. Wang, *On the number of short cycles in 5-connected planar triangulations*, arXiv doi10.48550/arXiv.2507.1809
 5. G. Agrahari and D. Froncek, *On Some Classes of Cycles-Related Γ -Harmonious Graphs*, *Utilitas Mathematica*, vol. 120, pp. 75–91, 2024, doi:10.61091/um120-07

Research Assistantship Experience

Summer Research Assistant	2026
Mentor: Zhiyu Wang, Yu Wang, Xiaonan Liu, LSU Department of Mathematics ¹ , Vanderbilt University ² <i>Project: Curvature-aware p-Laplacian-based graph neural networks</i>	
Summer Research Assistant	2024
Mentor: Nadejda Drenska ¹ , Peter Wolenski ¹ , and Steven Heymsfield ² LSU Department of Mathematics ¹ , Pennington Biomedical Research Center ² <i>Project: Predicting biometrics using supervised and semi-supervised learning</i>	

Summer Research Assistant
2023

Mentor: Peter Wolenski¹ and Yue Liu²
LSU Department of Mathematics ¹, Aquatic Germplasm and Genetic Resources Center²
Project: Predicting frog eggs count using deep learning

Teaching Experience (Instructor on Record)

Trigonometry	Spring 2026
Differential Calculus with Corequisite	Fall 2025
Differential Calculus	Spring 2025
College Algebra with Corequisite	Fall 2024
College Algebra	Fall 2023

Teaching Assistantship Experience

Grader, Linear Algebra	Spring 2025
Grader, Calculus III	Fall 2024
Grader, Calculus II	Spring 2024
Grader, Advanced Calculus	Spring 2024
Grader, Calculus III	Fall 2022
Lecture Assistant, Trigonometry	Fall 2022 & Spring 2023
Lab Tutor, Math Lab	Fall 2022, 2023, 2024 & Spring 2023,2025

Contributed Posters

Predicting ALM, BFP, and BMD using Semi-supervised Learning, <i>SIAM TX-LA Sectional Meeting, Austin TX</i>	September 2025
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Invited Talks

Predicting Body Biometrics using Semi-supervised Learning, <i>AMS Spring Southeastern Sectional Meeting, Clemson, SC</i>	March 2025
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Contributed Talks(LSU Seminars)

Seminar	Talk	Month/Year
LSU Combinatorics Semi-nar	Counting $K_{1,t}$ and $K_{2,t}$ in planar graphs with higher connectivity	April 2026
Combinatorics Student Seminar	Graph Labelings	April 2026
	Crapo’s Beta Invariant	February 2026
	The Necklace Problem	October 2025
AI Journal Club	How Graph Construction Im-pacts GNNs’ performances	March 2025

Contributed talks (Conferences)

Title	Conference	Location	Month/Year
Counting Subgraphs in Planar Graphs with Higher Connectivity	CombinaTexas	College Station, TX	April 2026
	Southeastern Int'l Conf. on Combinatorics, Graph Theory & Computing	Boca Raton, FL	March 2026
	AMS Southeastern Sectional Fall Meeting	New Orleans, LA	October 2025
Predicting Body Biometrics using Semi-supervised Learning	SIAM TX-LA Meeting 2025	Austin, TX	September 2025
	Scientific Computing Louisiana Around Louisiana	New Orleans, LA	March 2025
Graph Construction in Geospatial Analysis of EV Charging Stations	SIAM Conference on Mathematics of Data Science	Atlanta, GA	October 2024
On Some Classes of Cycles-related Harmonious Graphs	Math for All Conference	New Orleans, LA	April 2024
	Southeastern Int'l Conf. on Combinatorics, Graph Theory & Computing	Boca Raton, FL	March 2024
Colored Percolation in Three Dimensions	Eastern Great Lakes APS Meeting	Youngstown, OH	April 2022
Multiplicative Harmonious Labeling	Southeastern Int'l Conf. on Combinatorics, Graph Theory & Computing	Boca Raton, FL	March 2022
Alternative to Partial Fraction in Integration	Pi Mu Epsilon Conference	Youngstown, OH	February 2019
Economic Model for Environmental Cost of Developmental Projects	Pi Mu Epsilon Conference	Youngstown, OH	February 2019

Invited Summer Schools and Workshops

Distributed Open Education Network (DOENET) Workshop, <i>University of Minnesota, Twin Cities</i>	June 2026
Graduate Research Workshop in Combinatorics, <i>Iowa State University</i>	June 2025
Mathematical foundations of sampling connected balanced graph partitions, <i>American Institute of Mathematics</i>	June 2025
Topological Methods in Combinatorics, <i>Iowa State University</i>	May 2025

Other Workshops and Learning Communities

Virtual Workshop in Digital Accessibility <i>Mathematical Association of America, Ohio Section</i>	April 2026
2026 Virtual Summer Workshop Series,	May-August 2026

Center for Inquiry Based Learning, University of Michigan
Motivators Learning Community,
Motivate Lab, University of Virginia

2025-2026

Fellowships, Honors & Awards

Certificate of Teaching Excellence	Spring 2026
Arthur K. Barton Superior Graduate Student Scholarship	2025
Graduate Student Association Travel Award	2025
Corequisite Meaxutivator Fellowship	2025
David Oxley Graduate Student Teaching Award	Spring 2025
SIAM Travel Award, SIAM Mathematics of Data Science Conference	2024
Certificate of Teaching Excellence	Fall 2023
Meritorious Honor in COMAP's MCM/ICM	2022
Frank M Clark Physics Award	2021
Paul M Bancoczi Memorial Scholarship	2021
Meritorious Honor in COMAP's MCM/ICM	2019
YSU Honors College Scholarship	2018-2022
YSU International Scholar Award	2018-2022

Services

Co-organizer, Discussion and Training in Combinatorics	2025-2026
Session Co-organizer, Graphs and Networks, SIAM TX-LA Sectional Meeting	2025
Co-organizer, LSU Interdepartmental AI Seminar	2024-2025
Vice-President, SIAM LSU Chapter	2024-2025
Session Co-organizer, Graphs and Networks, SIAM Conference on MDS	2024
President, Nepalese Student Association, Youngstown State University	2019-2020

Volunteering

Session Assistant, Math Circle	March-April 2026
Research Panelist, LSU Math Graduate Student Orientation	August 2025
Campus Tour Guide, Prospective Graduate Students Visit	February 2025
Teaching Panelist, LSU Math Graduate Student Orientation	August 2024

Professional Membership

American Mathematical Society
Society for Industrial and Applied Mathematics
Mathematical Association of America

Mentorship Experience

Predicting Biometrics using Machine Learning

Spring, Summer 2024

- Mentored nine undergraduates and five graduates in collaboration with Pennington Biomedical Research Center, guiding students through the full research pipeline and achieving 90%+ accuracy in predicting biometrics such as lean mass and bone mineral density.
- Designed workshops on NumPy, supervised algorithms, and p-Laplacian methods; managed task allocation and timelines via a structured GitHub workflow.

Error-Correcting Codes Arising from Combinatorial Objects

Fall 2023

- Mentored two graduates and two undergraduates in researching error-correcting codes from combinatorial objects including projective planes, block designs, and Latin squares.
- Coordinated task distribution and milestones while guiding students in both theoretical development and Python implementation of cyclic linear codes and quotient rings of $x^n - 1$.

Graduate Projects

Resistance Curvature and Sprawling Graphs

January-May 2026

- Wrote a SageMath program to investigate the sprawling property of graphs. Co-proved the following results: Hamiltonian graphs, are sprawling, Sprawling graphs are 2-connected and are closed under edge addition.

Minimal Flag Triangulations of Torus

August-December 2025

- Contributed to the proof of the main result: The diameter of minimal flag triangulations of torus is between 2 and 4 by constructing several examples and investigating their graph properties.

Geospatial Equity Analysis of EV Charging Infrastructure : Dallas, TX

August 2025-

- Conducted multi-scale geospatial equity analysis of EV charging infrastructure in Dallas County, TX, combining OLS regression, spatial autocorrelation (Moran's $I = 0.299$), LISA cluster mapping, and spatial lag/error models at census tract and block group scales; identified 204 infrastructure-desert block groups invisible to tract-level policy tools and a $3 \times$ income-based equity gap across quintiles

Undergraduate Projects

Three-dimensional colored percolation (Undergraduate Thesis)

January-May 2022

- Designed a MATLAB program to calculate the percolation threshold of cubic lattices using the data collapsing technique within a two percent error of the known results. Extended conventional percolation systems to multi-colored using up to 100 colors

Portfolio Management, COMAP's MCM/ICM (Meritorious Honor)

February 2022

- Built an R-based LSTM time series model to predict gold and bitcoin prices, and designed a risk minimization algorithm with a client-facing recommendation system for optimal investment decisions

Environmental Cost of Development: An Economic Model, COMAP's MCM/ICM (Meritorious Honor)

January 2019

- Designed an ecological services valuation model to perform a cost-benefit analysis of land use developmental projects. Derived cost of lost land, land replacement, initial pollution and factory pollution for wetlands, forests, grassland and urban areas and concluded factory pollution is the biggest cost factor and projects on wetland have the highest environmental cost.

Internships

Session Assistant, Center For Talented Youth, Johns Hopkins University, Summer 2022

- Prepared evaluation reports on 23 high-school students across two sessions on their day-to-day progress and class behavior Organized math-focused evening sessions featuring fun, hands-on activities to reinforce key concepts.
- Advised students on crafting clear, structured, and professional solutions. Supervised the students on their final presentations on advanced topics in math logic and combinatorics.

Data Intern, Zoning Department, Government of Youngstown City Summer 2021

- Updated the zoning map using GIS as per the new regulations of the city. Studied the zoning applications to check the requirements before the approval of the zoning officer

Skills

Computational Tools: Python, SageMath, Mathematica, MATLAB, R, Excel, ArcGIS

Educational Tools: LaTeX, Doenet, PreTeXt, PreFigure